Specific Learning Objectives (SLOs)

Topic: General Anatomy of Bone (Skeletal System)

Phase I MBBS — Department of Anatomy, Murshidabad Medical College

Mapped to CBME competencies AN1.1, AN1.2, AN2.1–AN2.4

By the end of this class, using this handout for self-study, the student should be able to:

A. Introduction & Composition of Bone

Related competency code(s): AN1.2

SLO 1. Define bone and list its synonyms (os, osteon) and related terminology (osteology, ossification, osteomyelitis, osteomalacia).

SLO 2. Describe the composition of bone in terms of its organic and inorganic constituents and correlate composition with mechanical properties (hardness, resilience).

SLO 3. Describe the nature and distribution of red and yellow bone marrow, and its functional importance.

SLO 4. Enumerate the functions of bone (mechanical, protective, haemopoietic, mineral storage, immunological).

B. Classification of Bones

Related competency code(s): AN1.1 / AN2.3

SLO 5. Classify bones according to shape (long, short, flat, irregular, sesamoid, pneumatic) and give at least one example of each.

SLO 6. Classify bones according to their mode of development (membrane, cartilaginous, membrano-cartilaginous) with examples.

SLO 7. Classify bones according to region (axial and appendicular skeleton) and list the bones under each division.

SLO 8. Classify bones according to microscopic architecture (compact and cancellous; lamellar, woven, fibrous) and differentiate compact from cancellous bone.

SLO 9. Enumerate the special features and functions of a sesamoid bone.

SLO 10. Describe accessory (supernumerary) bones, give examples, and explain their medicolegal significance.

C. Gross Structure & Parts of a Long Bone

Related competency code(s): AN2.1

SLO 11. Describe the gross structure of an adult long bone under the headings periosteum, cortex, and medullary cavity.

SLO 12. Enumerate the parts of a young growing long bone (epiphysis, diaphysis, metaphysis, epiphyseal plate).

SLO 13. Classify epiphyses on the basis of number (simple, compound) and function (pressure, traction, atavistic, aberrant), with examples.

SLO 14. Explain the role of the epiphyseal plate of cartilage in the longitudinal growth of a bone.

D. Blood and Nerve Supply of Bone

Related competency code(s): AN2.1

SLO 15. Describe the sources of arterial supply of a growing long bone (nutrient, periosteal, epiphyseal, and metaphyseal arteries) and the territory each supplies.

SLO 16. Explain the clinical significance of the direction of the nutrient foramen in a long bone.

SLO 17. Describe the venous and lymphatic drainage of bone.

SLO 18. Describe the nerve supply of bone and its clinical relevance (periosteal sensitivity).

SLO 19. Correlate the vascular pattern at the metaphysis with the pathogenesis of osteomyelitis in children.

E. Development, Ossification & Growth of Bone

Related competency code(s): AN2.2

SLO 20. Differentiate between intramembranous (mesenchymal) and intracartilaginous (endochondral) ossification.

SLO 21. Define primary and secondary centres of ossification and describe their usual timing of appearance.

SLO 22. State the laws of ossification, including the recognised exception(s).

SLO 23. Describe how a long bone grows in length and in thickness, and define remodelling.

SLO 24. Enumerate the nutritional and hormonal factors that influence normal bone growth.

F. Medicolegal & Anthropological Aspects

Related competency code(s): AN1.2 (applied)

SLO 25. Enumerate the information a medicolegal expert must determine from a skeleton (species, number of individuals, age, sex, stature, race, time/cause of death).

SLO 26. Describe the basis of estimating skeletal age from the state of ossification and epiphyseal fusion.

SLO 27. Outline the basis of sex and stature estimation from bones.

G. Cartilage

Related competency code(s): AN2.4

SLO 28. Define cartilage and describe its general features (avascularity, absence of nerve supply, perichondrium).

SLO 29. Classify the types of cartilage (hyaline, fibrocartilage, elastic) and describe their characteristic locations in the body.

SLO 30. Differentiate between bone and cartilage with respect to structure, vascularity, nerve supply, and growth.

H. Applied/Clinical Anatomy of Bone

Related competency code(s): Integrated (Orthopaedics)

SLO 31. Correlate a defect in intramembranous ossification with cleidocranial dysostosis, and a defect in endochondral ossification with achondroplasia.

SLO 32. Explain the anatomical basis of pain in periosteal disease and the general principles of fracture healing.

SLO 33. Correlate deficiency of vitamin D/calcium in children and adults with rickets and osteomalacia respectively, and differentiate osteoporosis from osteomalacia.

SLO 34. Identify common sites used for bone marrow biopsy and correlate with the distribution of red marrow in the adult skeleton.

Note on Use

These Specific Learning Objectives (SLOs) break down the broad CBME competencies for the topic "Skeleton" into smaller, assessable statements. Each SLO is addressed in the accompanying self-learning handout, and is tested through the attached MCQs and explanation-based questions. Students are encouraged to self-check against this list after completing the handout.